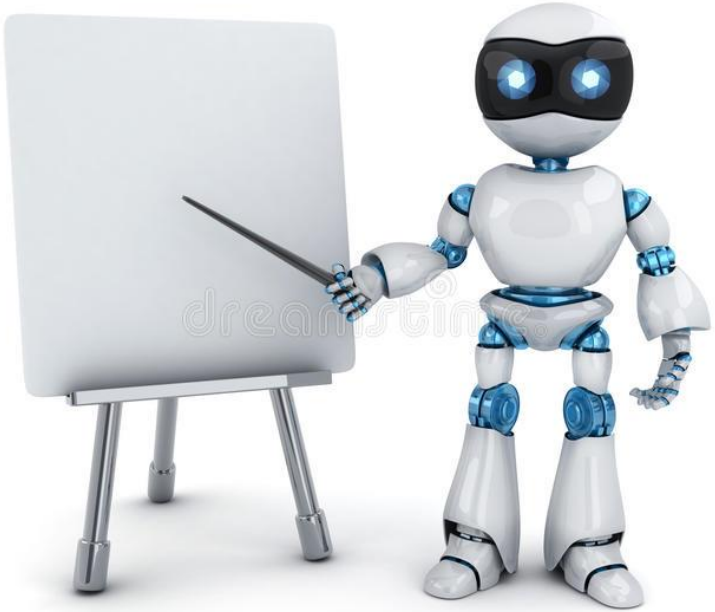

LINUX

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WHAT IS LINUX?

Linux is an operating system which sits in the middle of your hardware and users. It helps hardware to communicate with software.

Hardware

Operating system

MacOS

Windows

Linux

Software

Google

YouTube

python™

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WHO CREATED LINUX?

- Linux is an operating system or a kernel which germinated as an idea in the mind of young and bright **Linus Torvalds** when he was a computer science student.
- He has launched **LINUX OS in 1991** when his ideas were rejected by the designers of **UNIX OS.**
- Now, Linux is the most popular and widely used kernel because it is open source and offers variety of functions to the users.

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WHY LINUX?

- It offers free operating system.
- It is easy to learn Linux for beginners.
- Linux operating system offers millions of applications and Linux software for free.
- Linux is highly secure system once you have installed you no longer need an antivirus.
- To prevent hacking attempts, many organizations keep their Linux operating system private.
- Mega – companies like Amazon , Facebook and Google use Linux for their servers because Linux based server can run non-stop without a reboot for years on end.

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LINUX USERS

- US Government and Agencies
- NASA
- Health care
- Traffic control
- Bullet trains in Japan run at the speed of 150-215 m/h.
- Entertainment industries
- World e-commerce leaders including Amazon , eBay ,PayPal and Walmart run their platform on Linux.
- Other fortune 500 companies such as Google , IBM , McDonalds , Facebook uses Linux.

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WT IS LINUX DISTRIBUTION?

- Linux distribution often shortened to “Linux distros” is a version of Linux operating system that is packaged with other components such as an installation programs, management tools.
- Each Linux distribution is built for specific purpose to meet the demands of its target users.
- There are hundreds of Linux distributions are available and each targets specific users or systems such as desktops , servers, mobile devices or embedded devices.
- Linux distribution consists of software packages that contain files , applications or services.
- Linux distribution also includes package management system which is used to install , uninstall and manage software packages.

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LINUX DISTROS

fedora

ubuntu

SUSE

CentOS

Red Hat

debian

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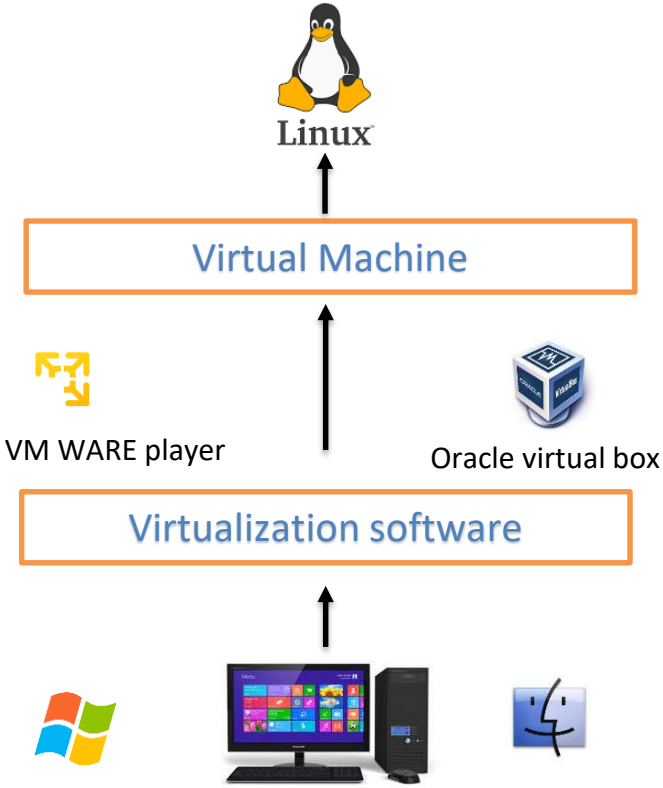
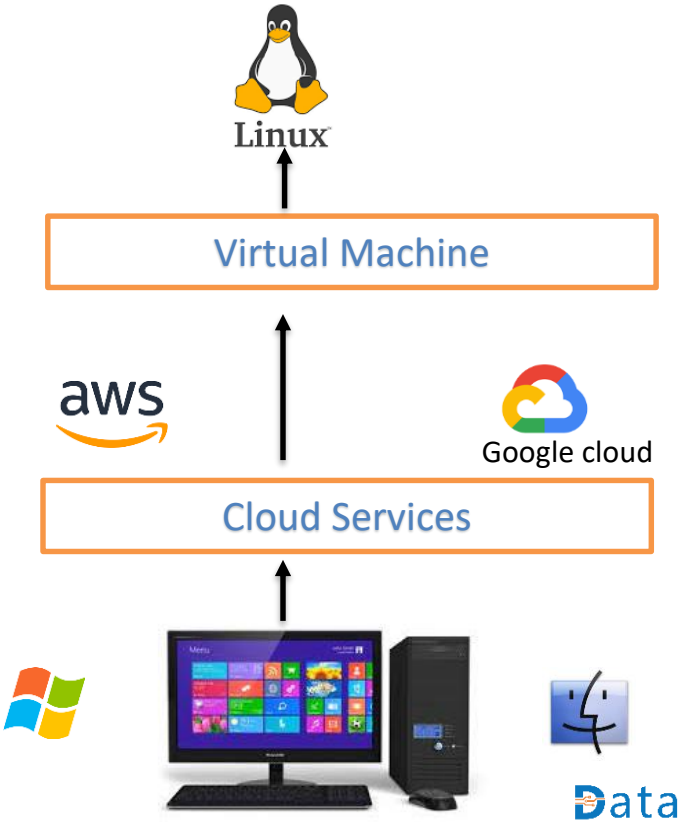
WAYS TO INSTALL LINUX OS





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FILE MANAGEMENT IN LINUX

- Just like in every other operating system like Windows and Mac OS we can control and manage files like creating, removing deleting, copying, cutting, and pasting using the GUI.
- In Linux, files are arranged in a hierarchical structure of directories or folders.
- A directory is simply a container for your files. Directories can further have subdirectories.
- The Linux file structure spans in a manner similar to branches of a tree and is hence commonly referred to as a tree structure. It begins with a **root** directory, which is referenced using **/**, or forward slash, on the terminal.
- The **root** directory contains a number of system directories and files that govern your Linux system.

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FILE ATTRIBUTES

- **Name:** It is the only information stored in a human-readable form.
- **Identifier**: Every file is identified by a unique tag number within a file system known as an identifier.
- **Location:** Points to file location on device.
- **Type:** This attribute is required for systems that support various types of files.
- **Size**. Attribute used to display the current file size.
- **Protection**. This attribute assigns and controls the access rights of reading, writing, and executing the file.
- **Time, date and security:** It is used for protection, security, and also used for monitoring

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FILE TYPES

Linux has 3 types of files:

- **Regular files**

It is the common file type in Linux. it includes files like – text files, images, binary files, etc.

- **Directories**

Windows call these directories as folders. These are the files that store the list of file names and the related information.

- **Special files**

Represents a real physical device such as a printer which is used for IO operations. Device or special files are used for device Input/Output(I/O) on UNIX and Linux systems.

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CREATING FILES

The **touch** command is used to create files and directories.

Syntax

$ touch filename

Create new file hello.txt

$ touch hello.txt

Create new file in another location

$ touch /home/Python/hello.txt

Add content to the file hello.txt as "Data Science is booming"

$ cat > hello.txt

Data science is booming

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COPY FILES

The **cp** command is used to copy files into current directory or to some other directory.

Syntax

$ cp filename destination

Copy text.pdf to Python directory

$ cp text.pdf Python/

Copy Python directory to documents directory

$ cp –r Python/Documents/

Copy content from txt1.pdf to txt2.pdf

$ cp tx1.pdf txt2.pdf

Copy content of multiple files into single file using cat **command**

$ cat file1.txt file2.txt file3.txt

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MOVE FILES

The **mv** command is used to move a file into another directory. It can also be used to rename file or directory.

Syntax

$ mv filename destination

Move sample.txt to python directory

$ mv sample.txt python/

Move Python directory to documents directory

$ mv Python/ Documents

Move multiple files into python directory

$ mv hello1.txt hello2.txt hello3.txt Python/

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DELETE FILES

The **rm** command is used to remove files and directories.

Syntax

$ rm filename

Remove sample.txt file

$ rm sample.txt

Remove Python directory

$ rm –r Python/

Remove multiple files

$ rm hello1.txt hello2.txt hello3.txt

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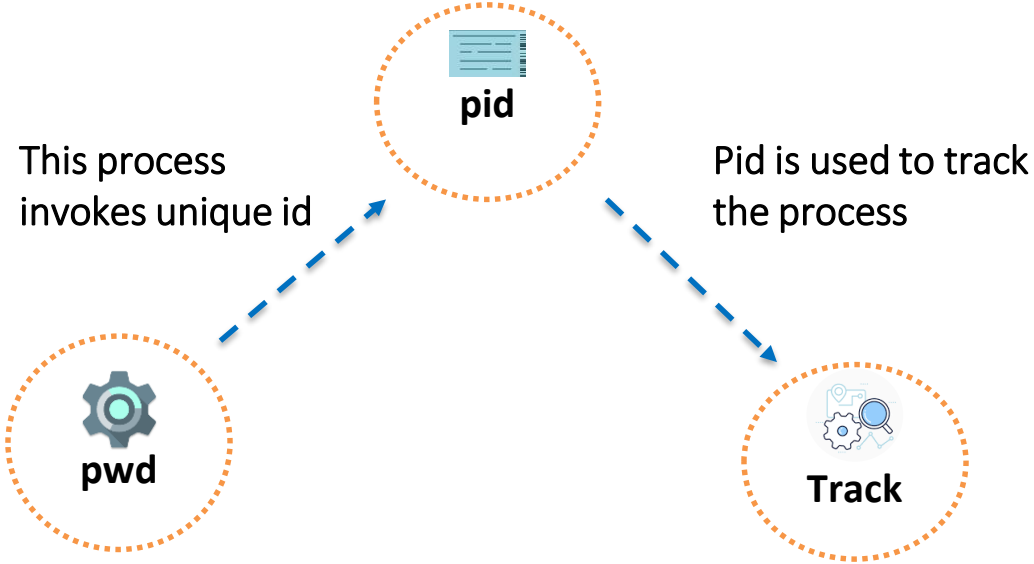
PROCESS CONTROL COMMANDS

An instance of a program is called a Process. In simple terms, any command that you give to your Linux machine starts a new process. It invokes the 5 digit unique id called **pid**. Using pid , the process can be tracked.



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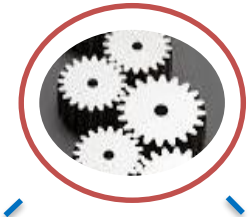
TYPES OF PROCESS

A process can be initialized in two ways, foreground and background.



Foreground

They run on the screen by default and need input from the user.
Eg: Office programs

Background

They run in the background and allows other process to run in parallel.
Eg: Antivirus

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FOREGROUND PROCESS

- To start a foreground process, you can either run it from the dashboard, or you can run it from the terminal.
- When using the Terminal, you will have to wait, until the foreground process runs.

OR

```
home@VirtualBox:~$ banshee
```


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FOREGROUND PROCESS

- If you start a foreground program/process from the terminal, then you cannot work on the terminal, till the program is up and running.
- Particular, data-intensive tasks take lots of processing power and may even take hours to complete. You do not want your terminal to be held up for such a long time.
- To avoid such a situation, you can run the program and send it to the background so that terminal remains available to you.

Start the program and press ctrl+z

```
guru99@VirtualBox:~$ banshee
[Info  16:08:36.688] Running Banshee 2.2.1: [Ubuntu 11.
11-12-19 14:51:26 UTC]
^Z
[1]+  Stopped                 banshee
```

Type 'bg' to send the process to the background

```
guru99@VirtualBox:~$ bg
```

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top

The **top** command is used to track all the running process within the working environment of Linux system.

Syntax

$ top

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kill

- In order to stop a **foreground** running process use the following command

$ ctrl +c

- The **kill** command in Linux is used to stop a **background** running process.
- We need to know the PID (process id) of the process we want to kill.

Syntax

$ kill PID

How to find pid of process name

$ pidof processname

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ps

- The **ps** command in Linux is used to list all the running processes.
- This command stands for 'Process Status'. It is similar to the "Task Manager" that pop-ups in a Windows Machine when we use Ctrl+Alt+Del. This command is similar to 'top' command but the information displayed is different.

Syntax

$ ps

To check all the process running under user

$ ps –u **or** $ ps-u username

To check process status of single id

$ ps PID

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free

This **free** command shows the free and used memory (RAM) on the Linux system.

Syntax

$ free

$ free -m → displays output in MB

$ free –g → displays output in GB

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df

This utility reports the free disk space(Hard Disk) on all the file systems.

Syntax

$ df

If you want the above information in a readable format, then use the command

$ df -h

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Nice

- Linux can run a lot of processes at a time, which can slow down the speed of some high priority processes and result in poor performance.
- To avoid this, you can tell your machine to prioritize processes as per your requirements.
- This priority is called Niceness in Linux, and it has a value between -20 to 19. The lower the Niceness index, the higher would be a priority given to that task.
- The default value of all the processes is 0.

Syntax

$ nice -n [nice value] [process]

$ nice -n 19 google

If there is some process already running on the system, then you can 'Renice' its value using syntax.

$ renice [nice_value] -p [PID]

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FILE PERMISSION

- Linux is the multi-user operating system which can be accessed by many users simultaneously.
- Linux can also be used in mainframes and servers without any modifications. But this raises security concerns as an unsolicited or malign user can corrupt, change or remove crucial data.

For effective security, Linux divides authorization into 2 levels.

1. Ownership
2. Permission

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OWNERSHIP OF LINUX FILES

User

Groups

Others/processes

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GROUPS

- A user- group can contain multiple users and all users have same Linux group permission access to the file.
- Suppose you have a project where a number of people require access to a files.
- Instead of manually assigning permissions to each user, you could add all users to a group, and assign group permission to file such that only this group members and no one else can read or modify the files.

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OTHERS

- Any other user who has access to a file.
- This person has neither created the file, nor he belongs to a user group who could own the file.
- Practically, it means everybody else. Hence, when you set the permission for others, it is also referred as set permissions for the world.

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USER

- A user is the owner of the file. By default, the person who created a file becomes its owner. Hence, a user is also sometimes called an owner.
- System uses User IDs to manage users.

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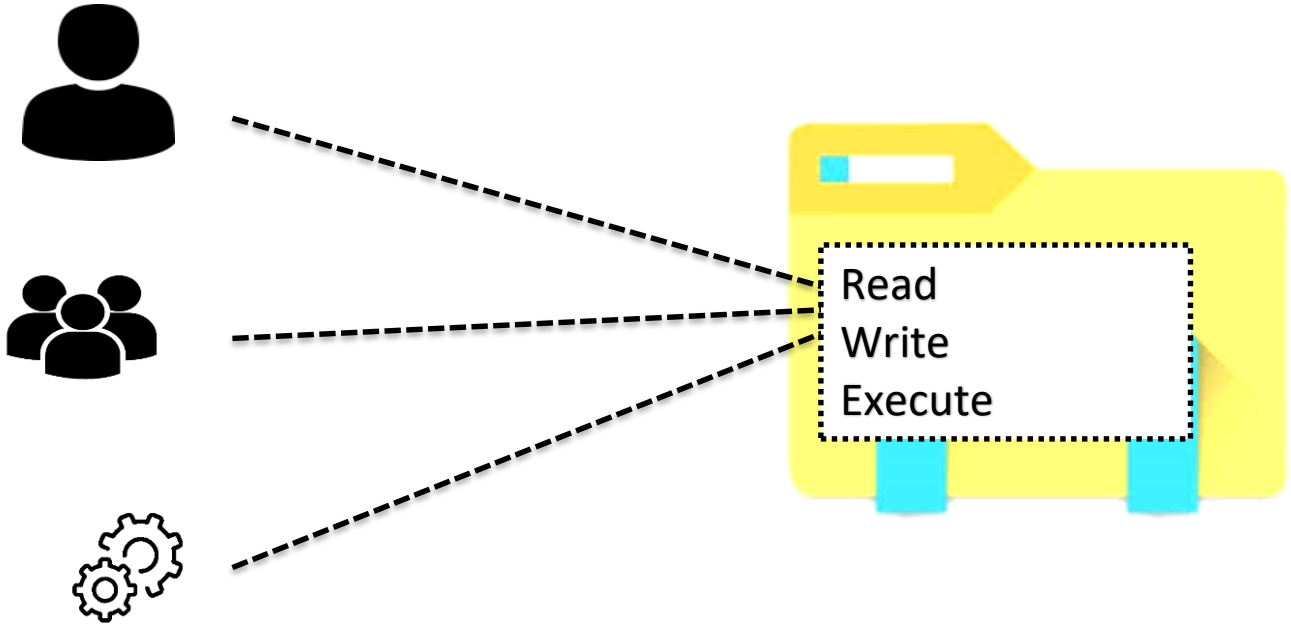
LINUX FILE PERMISSION



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LINUX FILE PERMISSION

Read

This permission give you the authority to open and read a file. Read permission on a directory gives you the ability to lists its content.

Write

The write permission gives you the authority to modify the contents of a file. The write permission on a directory gives you the authority to add, remove and rename files stored in the directory.

Execute

In Windows, an executable program usually has an extension ".exe" and which you can easily run. In Unix/Linux, you cannot run a program unless the execute permission is set. If the execute permission is not set, you might still be able to see/modify the program code(provided read & write permissions are set), but not run it.

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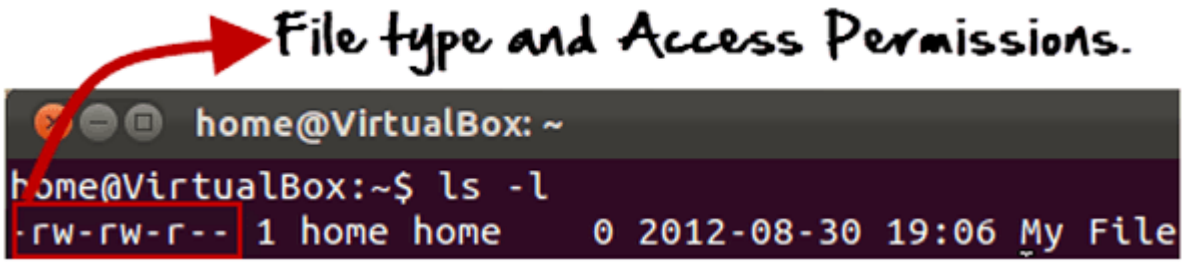
FILE TYPE AND ACCESS PERMISSIONS

The **ls -l** command is used to check file type and access permission.



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FILE TYPE AND ACCESS PERMISSIONS

The highlighted **'-rw-rw-r-'**and this weird looking code is the one that tells us about the Unix permissions given to the owner, user group and the world.

```
-rw-rw-r--
```

indicates file

d represents directory

```
drwxr-xr-x 2 ubuntu ubuntu 80 Sep  6 07:27 Desktop
```

r = read permission
w = write permission
x = execute permission
– = no permission
d = directory

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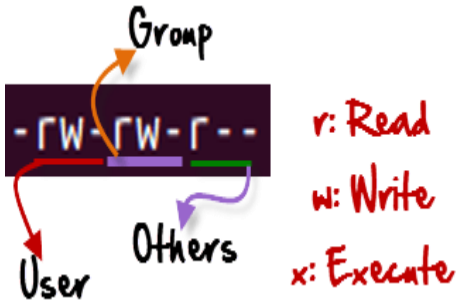
FILE TYPE AND ACCESS PERMISSIONS

The first part of the code is ‘rw-‘. This suggests that the owner ‘Home’ can:

- Read the file
- Write or edit the file
- He cannot execute the file since the execute bit is set to ‘-‘.



The second part is ‘rw-‘. It for the user group ‘Home’ and group-members can:

- Read the file
- Write or edit the file

The third part is for the world which means any user. It says ‘r–‘. This means the user can only:

- Read the file

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CHANGING FILE OR DIRECTORIES

The ‘**chmod’** command which stands for ‘change mode’. Using this command, we can set permissions (read, write, execute) on a file/directory for the owner, group and the world.

Syntax

$ chmod permission filename

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CHANGING FILE OR DIRECTORIES

Permissions can be defined in two ways

- Absolute
- Symbolic

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ABSOLUTE MODE(NUMERIC)

In **absolute mode**, file permissions are not represented as characters but **a three-digit octal number.**

Number	Permission type	Symbol
0	No permission	-
1	Execute	-x
2	Write	-w-
3	Execute+write	-wx
4	Read	r-
5	Read+execute	r-x
6	Read+write	rw-
7	Read+write+execute	rwx

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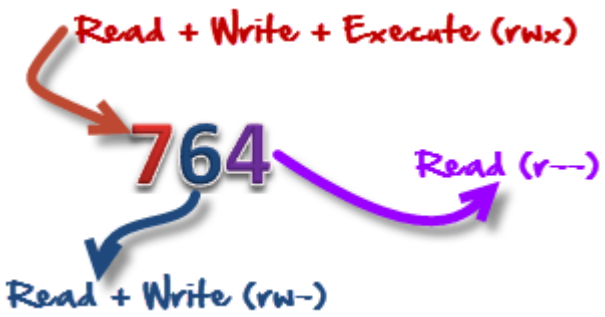
EXAMPLE

Let us check permission of the sample file using **ls –l** command.

```
ubuntu@ubuntu:~$ ls -l sample
-rw-rw-r-- 1 ubuntu ubuntu 15 Sep  6 08:00 sample
```

Let us change permission of the file using **absolute chmod** command.

```
ubuntu@ubuntu:~$ chmod 764 sample
ubuntu@ubuntu:~$ ls -l sample
-rwxrw-r-- 1 ubuntu ubuntu 15 Sep  6 08:00 sample
```



764 absolute code says the following:

- Owner can read, write and execute
- Usergroup can read and write
- World can only read

This is shown as '-rwxrw-r–

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SYMBOLIC MODE(NUMERIC)

In **symbolic mode** we can modify permissions of a specific owner. It uses mathematical symbols to modify file permissions.

operator	Description	Symbol	members
+	Adds permission to file or directory	u	User/owner
-	Removes the permission	g	Group
=	Sets the permission and overrides the permissions set earlier	o	Others
		a	all

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EXAMPLE

Here we will be using characters like **rwx** to set permission.

Current File Permissions

```
home@VirtualBox:~$ ls -l sample
-rw-rw-r-- 1 home home 55 2012-09-10 10:59 sample
```

Setting permissions to the 'other' users

```
home@VirtualBox:~$ chmod o=rwx sample
home@VirtualBox:~$ ls -l sample
-rw-rw-rwx 1 home home 55 2012-09-10 10:59 sample
```

Adding 'execute' permission to the usergroup

```
home@VirtualBox:~$ chmod g+x sample
home@VirtualBox:~$ ls -l sample
-rw-rwxrwx 1 home home 55 2012-09-10 10:59 sample
```

Removing 'read' permission for 'user'

```
home@VirtualBox:~$ chmod u-r sample
home@VirtualBox:~$ ls -l sample
--w-rwxrwx 1 home home 55 2012-09-10 10:59 sample
```


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ls

- The **ls** command in linux is used to display a directory's files and folders.

SYNTAX:

$ ls

- The **ls –l** is used to display a detailed version of every file and directory available inside the current directory.

SYNTAX:

$ ls -l

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cat

- **$cat <filenames>** command in Linus is used to read the contents of one or more files and display their contents.

SYNTAX:

$ cat hello.txt

$ cat hello1.txt hello2.txt

- **$Cat –n <filenames>** is used to display content of file along with line numbers.

SYNTAX:

$ cat -n hello.txt

$ cat -n hello1.txt hello2.txt

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cd

- The **cd** command in Linux is used to change current working directory to a specified folder.

SYNTAX:

The cd command is used to navigate to the Linux folder inside our Desktop directory.

$ cd Desktop/Linux/

This command is used to navigate to the parent directory of the current directory which Desktop.

$ cd ..

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pwd

- The **pwd** command in Linux is used to display the path of the current working directory.

SYNTAX:

$ pwd

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man

- The **man** command in Linux is used to display the documentation/ user manual. It includes the name of the bash command , its detailed synopsis , a short description, existing versions of the command as well as authors of the bash command.

SYNTAX:

Displays documentation on the ls command.

$ man ls

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touch

- The **touch** command is used to create a new files without any content inside in it.

SYNTAX:

Displays documentation on the ls command.

$ touch <filenames>

$ touch text1.txt

$ touch text1.css text2.html

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mkdir

- The **mkdir** command is used to create a new directories inside an existing working directory from the terminal.

SYNTAX:

Used to display relevant information pertaining to the **mkdir** command.

$ mkdir –help

Created python folder using mkdir

$ mkdir Python

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echo

The **echo** command in Linux simply displays a line of text/string which is passed in as an argument. It is commonly used for debugging shell programs inside the terminal.

SYNTAX:

Displays python

$ echo 'Python'

Displays "Learn Python from Datamites" , **'-e'** tag used to recognize the backslash escape sequence inside the argument.

$ echo –e "Learn \nPython \nfrom \nDatamites"

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rm

The **rm** command in Linux is used to delete files and directories.

SYNTAX:

Delete single file

$ rm hello.txt

The '-i ' tag enables rm command to request confirmation from the user before deleting corresponding file.

$ echo –e "Learn \nPython \nfrom \nDatamites"

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rmdir

The **rmdir** command in Linux is used to delete empty directories. If a directory has certain files/folders inside it then it will display an error.

SYNTAX:

$ rmdir –help

$ rmdir Linux/

$ rmdir Python/

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wget

The **wget** command in Linux is a utility tool to download applications/ web pages directly from the web. It allows file downloads with HTTP , HTT and FTP protocols.

SYNTAX:

$ wget <url>

$ wget https://html.com/

The '-O' is used to retry a file download for a given number of times.

$ wget -O <filename> <url>

$ wget –O webpage.html https://html.com/

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mv

The **mv** command in Linux translates to move. It performs two major functions in Linux.

- Used to rename file or directory.

SYNTAX:

$ mv <old_name> <new_name>

- Used to move a file or directory from one location to another.

SYNTAX:

Move index.html to python folder

$ mv index.html Python/

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cp

The **cp** command in Linux translates to copy. It is used to copy files/directories from one location to another. It is used to copy contents of source file into the destination file.

SYNTAX:

$ cp <source_loc> <dest_loc>

$ cp <source_file> <dest_file>

Content of text1 is transferred/overwritten to txt2

$ cp text1.txt text2.txt

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tree

The **tree** command in Linux is used to list out the contents of directories in a tree like fashion.

SYNTAX:

$ tree

$ tree –f displays full path of each working directory and file inside current directory.

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grep

The **grep** command in Linux searches through a specified file and prints all lines that match a given pattern.

SYNTAX:

$ grep <option> <pattern> <filename>

$ grep 'Datamites' hello.txt

The '-i' tag used to disable case sensitivity.

$ grep -i 'Datamites' hello.txt

The '-c' tag used to count the number of occurrences of a given pattern inside a file.

$ grep -c 'Datamites' hello.txt

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history

The **history** command in Linux used to view a history of all the commands previously executed inside the bash terminal.

SYNTAX:

$ history

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df

The **df** command in Linux is used to display information related to file system about total space and available space. If file name is not given , it displays the space available on all currently mounted file systems.

SYNTAX:

$ df

$ df –a

$ df –total

$ df –T /home/python

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rename

The **rename** command in Linux is used to rename files. It is useful in renaming large group of files.

SYNTAX:

$ sudo apt install rename

 # rename character in filename

$ rename ‘y/abc/xyz/’

In this example , every a character in the filename is replaced by an x, every b by y, and every c by z.

 # change file extension txt to pdf

$ rename –v ‘s/.txt/.pdf/ ’*.txt

 # replacing part of filename example.txt to instance.txt

$ rename –v ‘s/example/test/ ’*.txt

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locate

The **locate** command in Linux is used to find files in Linux using the file name

SYNTAX:

$ locate

Search file with specific name

$ locate python.py

List 20 files containing html pattern in the filename.

$ locate "*.html" –n 20

The '-c ' tag is used to find number of files containing mysql string in filename

$ locate –c [.txt]*

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top

The **find** command in Linux is used to show the Linux process.

SYNTAX:

$ find

Find file by name

$ find –name “quer

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find

The **find** command in Linux is used to find files and files in directory.

SYNTAX:

$ find

Find file by name

$ find . –name hello1.txt

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zip

The **zip** command is a compression and file packaging utility for linux. Each file stored with zip extension.

```
$ sudo apt install zip
```

Zip single file

```
$ zip test.zip testfile.txt
```

Zip Multiple files

```
$ sudo touch testfile1.txt testfile2.txt testfile3.txt testfile4.txt
$ zip files.zip testfile1.txt testfile2.txt testfile3.txt testfile4.txt
```

Delete original files after zip

```
$ zip -m test.zip testfile.txt
```


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unzip

The **unzip** command in Linux is used to extract files from zip file.

```
$ sudo apt install unzip
```

Zip single file

```
$ unzip test.zip
```

Unzip files to other directory

```
$ unzip test.zip -d Documents/
```

Unzip multiple files

It will unzip all the files in current directory.

```
$ unzip '*.zip'
```

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uname

The **uname** command in Linux displays all the information about system.

Syntax:

$ uname –a → prints system information in the following order:kernel name, host name , release date , version , hardware name, hardware platform, operating system.

$ uname –s → prints kernel name

$ uname –n →prints host name

$ uname –r →prints release date

$ uname –v → prints version

$ uname –m → prints machine hardware

$ uname –p →print type of processor

$ uname –i → prints platform of hardware

$ uname –o →prints name of the operating system.

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hostname

The **hostname** command in Linux is used to display hostname and to set host name.

Syntax:

$ hostname

$ hostname -a → used to get alias name of host system.

$ hostname -b → used to set hostname

$ hostname -i → used to get IP address

$ hostman -s → used to get hostname in short

$ hostman -V → used to get version number

$ sudo hostman new_name → used to set hostname

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hostname

The **hostname** command in Linux is used to set hostname and to display hostname of the system

$ sudo apt install unzip

Zip single file

$ unzip test.zip

Unzip files to other directory

$ unzip test.zip -d Documents/

Unzip multiple files

It will unzip all the files in current directory.

$ unzip '*.zip'

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useradd

The **useradd** is a command in Linux that is used to add user accounts to your system.

Syntax

$ useradd

Add simple user

$ sudo useradd test_user

Create a user with specific user id

$ sudo useradd –u 1234 test_user

Create a user with expiry date

$ sudo useradd –e 2023-03-30 test_user

Create user with changed login shell

$ sudo useradd –s /bin/sh test_user

Set an unencrypted password for the user

$ sudo useradd –p test_password test_user

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sudo

- Super user Do (sudo) command in Linux is generally used as a prefix of some command that only superusers are allowed to run. It allows users with proper permissions to execute a command as another user , such as superuser. The option of sudo let us have multiple administrators and is same as “run as administrator” option in windows.

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NETWORKING IN LINUX

- A group of interconnected computers are called a Network and working on netwoks and handling them is called as Networking.
- A system administrator’s routine task include configuring , maintaining , trouble shooting and managing servers and networks within data centers.

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WHY NETWORKING?

- Resources can be shared across the network.
- We can communicate across the network with other users

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NETWORKING COMMANDS

- Ifconfig
- Ping
- Wget
- Curl
- Ssh
- Scp
- ftp

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NETWORKING COMMANDS

The **ifconfig** command is used to configure how kernel links up the software side to the hardware side using a network interface.

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SHELL SCRIPTING

If you are using any major operating system you are indirectly interacting to **shell**. If you are running Ubuntu, Linux Mint or any other Linux distribution, you are interacting to shell every time you use terminal.

Before understanding shell scripting we have to get familiar with following terminologies:

- Kernel
- Shell
- Terminal

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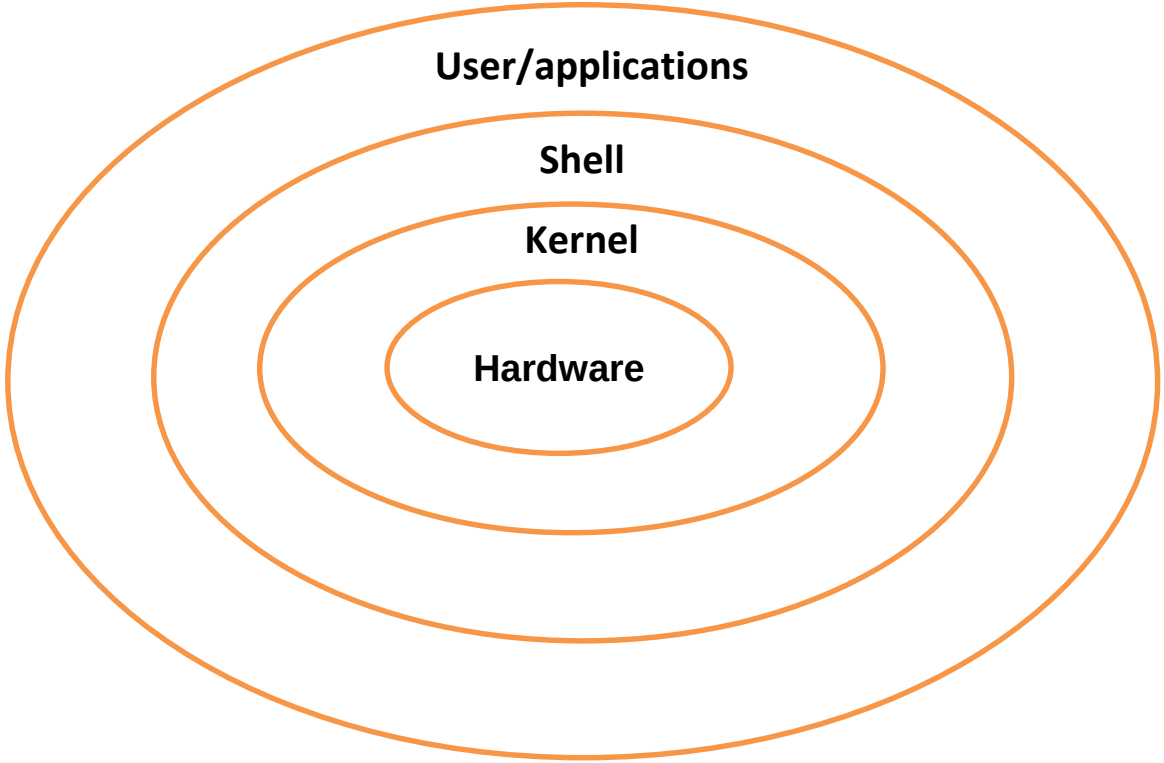
SHELL SCRIPTING



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WHAT IS KERNEL?

The kernel is a computer program that is the core of a computer's operating system, with complete control over everything in the system.

It manages following resources of the Linux system –

- File management
- Process management
- I/O management
- Memory management
- Device management etc.

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WHAT IS SHELL?

- A shell is special user program which provide an interface to user to use operating system services.
- Shell accept human readable commands from user and convert them into something which kernel can understand.
- It is a command language interpreter that execute commands read from input devices such as keyboards or from files.
- The shell gets started when the user logs in or start the terminal.
- It will check whether the command is valid or not.

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SHELL TYPES

- BASH (Bourne Again SHell) – It is most widely used shell in Linux systems. It is used as default login shell in Linux systems and in macOS. It can also be installed on Windows OS.
- CSH (C SHell) – The C shell's syntax and usage are very similar to the C programming language.
- KSH (Korn SHell) – The Korn Shell also was the base for the POSIX Shell standard specifications etc.

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SHELL SCRIPTING

- Usually shells are interactive that mean, they accept command as input from users and execute them.
- However some time we want to execute a bunch of commands in that case we can write these commands in a file and can execute them in shell to avoid this repetitive work. These files are called **Shell Scripts** or **Shell Programs**.
- Shell scripts are similar to the batch file in MS-DOS. Each shell script is saved with **.sh** file extension eg. **myscript.sh**
- A shell script have syntax just like any other programming language. If you have any prior experience with any programming language like Python, C/C++ etc. it would be very easy to get started with it.

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WHY SHELL SCRIPT?

There are many reasons to write shell scripts:

- To avoid repetitive work and automation
- System admins use shell scripting for routine backups
- System monitoring
- Adding new functionality to the shell etc.

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ADVANTAGES OF SHELL SCRIPT

- The command and syntax are exactly the same as those directly entered in command line, so programmer do not need to switch to entirely different syntax
- Writing shell scripts are much quicker
- Quick start
- Interactive debugging etc.

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DISADVANTAGES OF SHELL SCRIPT

- Prone to costly errors, a single mistake can change the command which might be harmful
- Slow execution speed
- Design flaws within the language syntax or implementation
- Not well suited for large and complex task
- Provide minimal data structure unlike other scripting languages. etc

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HOW SHELL SCRIPTS WORKS?

Create folder

$ pwd

$ mkdir Scripts

$ cd Scripts

Create file

$ gedit test.sh

Ctrl S+ctsl Q

Give execute permission

$ chmod u+x test.sh

$ls –l test.sh

Run the file

$ ./test.sh

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SHELL SCRIPT ELEMENTS

A shell script comprises following elements –

- Shell Keywords – if, else, break etc.
- Shell commands – cd, ls, echo, pwd, touch etc.
- Functions
- Control flow – if..then..else, case and shell loops etc.

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SHELL VARIABLES

- The name of a variable can contain only letters (a to z or A to Z), numbers (0 to 9) or the underscore character (_) and cannot use other characters such as **!**, *****, or **-** is that these characters have a special meaning for the shell.

Define variable

```
Course="python"
Value=100
```

Accessing variables

```
course='python'
echo $course
```

- Echo is used to print the output.

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ARITHMETIC AND COMPARISON OPERATORS

Operator
+ (Addition)
- (Subtraction)
* (Multiplication)
/ (Division)
% (Modulus)

Operator	Example
-eq → equal	[$a -eq $b]
-ne → not equal	[$a -ne $b]
-gt → greater than	[$a -gt $b]
-lt → less than	[$a -lt $b]
-ge → greater than or equal to	[$a -ge $b]
-le → less than or equal to	[$a -le $b]

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Logical Operators

Operator	Logical	Expression
-o	or	[$a -lt 20 -o $b -gt 100]
-a	and	[$a -lt 20 -a $b -gt 100]
!	not	[! false]

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CONDITIONAL STATEMENTS

Linux Shell supports conditional statements which are used to perform different actions based on different conditions.

The two decision-making statements here −

1)**The if...else statement**

- if…fi
- If…else…fi
- If…elif..else..fi

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IF

Syntax:

```
if [condition]
then
    statements
fi
```

Eg:: Check whether a is less than b or not

```
a=20
b=30
if [ $a –lt $b]
then
    echo “ $a is greater than $b”
fi
```


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IF-ELSE

Syntax:

```
if [condition]
then
    statements
else
   statements
fi
```

Eg:: Check whether a is less than b or not

```
a=200
b=30
if [ $a –lt $b]
then
    echo “ $a is greater than $b”
else
    echo “$ a is less than $b”
fi
```


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IF-ELIF-ELSE

Syntax:

```
if [condition]
then
    statements
elif [condition]
then
    statements
elif [condition]
then
    statements
else
   statements
fi
```

Syntax:

```
a=10
b=20
if [ $a –eq $b ]
then
    echo “a is b”
elif [ $a –gt $b ]
then
    echo “ a is greater than b”
elif [ $b % $a –eq 1 ]
then
    echo “ remainder is one”
else
   echo “a is less than b”
fi
```

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WHILE

Syntax:

```
while [condition]
do
    statements
done
```


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FOR

Syntax:

```
for var in <list>
do
   statements
done
```

```
Name="Jeevitha"
for var in $Name
do
   echo $var
Done
```

Range

```
for var in { 1..5 }
do
   echo $var
Done
```

Range with step

```
for var in { 10..0..2 }
do
   echo $var
Done
```

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FUNCTIONS

Syntax
{ **function_name**} ( )
{
    statements
}
# call function
{**function name**}

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SCHEDULING TASK IN LINUX

- Whenever using a UNIX-based operating system, certain tasks are to be performed repeatedly.
- Running them manually every single time is time-consuming and overall inefficient.
- To solve this issue, UNIX comes with its built-in task schedulers such as **cron and crontab.**
- These task schedulers act like a smart alarm clock. When the alarm goes off, the operating system will run the predefined task.
- For example, you could schedule a particular job to periodically execute at specific times of the day. This is helpful for performing daily backups, monthly log archiving, weekly file deletion to create space, and so on.

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WHAT IS CRON?

- The **cron** daemon is responsible for running a lot of jobs at specific times in the background.
- The crond daemon enables cron functionality and runs in background. The cron reads the **crontab** (cron tables) for running predefined scripts.
- It offers great flexibility irrespective of the task, irrespective of the interval (hour, week, month, year, or whatever).
- Cron keeps track of its actions using the crontab file.
- The crontab file is a script that contains all the necessary information to run all the cron jobs.

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WHY CRONJOBS?

- Helps OS to take a scheduled backup of log files or database.
- Delete old log files
- Send out any notification email such as Newsletters, Password expiration email
- Regular clean-up of cached data
- Crontab is an ideal option to automate Unix jobs.
- It is used to automate system maintenance

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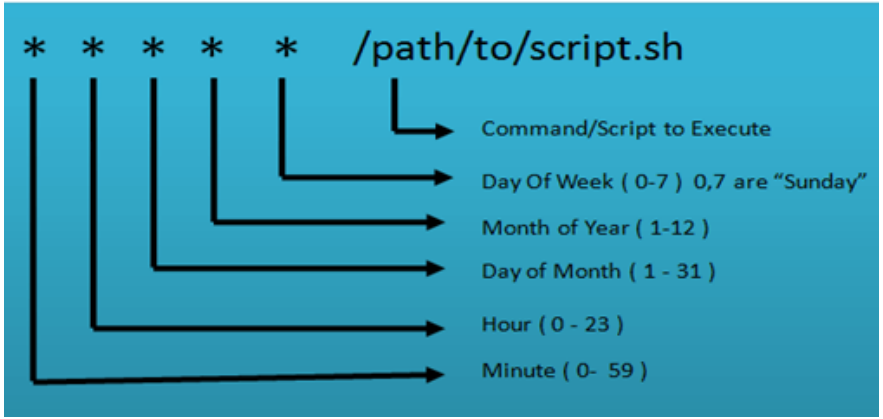
CRONTAB SYNTAX

SYNTAX
[Minute]
[hour]
[Day_of_the_Month]
[Month_of_the_Year]
[Day_of_the_Week]
[command]



•**Astrics (*):** Use for matching
•**Define range:** Allows you to define a range with the help of hyphen like 1-10 or 30-40 or jan-mar, mon-wed.
•**Define multiple ranges:** Allows you to define various ranges with command separated like apr-jun,oct-dec.

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TYPES OF ACCOUNTS IN LINUX

- **Root /administrative**

The Linux administrative root account is automatically created when you install Linux , and it has administrative privileges for all the services on Linux operating system. The root account is known as **super user.**

- **Regular**

Regular users have the necessary privileges to perform standard tasks on a Linux computer such as running word processors , databases and web browsers. They can store files in directory but they cannot delete critical operating system configuration files.

- **Service**

Services such as Apache , Squid , mail , games and printing have their own individual service accounts. These accounts exist to allow each of these services to interact with your computer

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USER MANAGEMENT

- As Linux is multi-user operating system , there is a high need of an administrator, who can manage user accounts and their rights.
- User management includes everything from creating a user to deleting user on your system.
- Each user is assigned an ID that is unique for each user in the operating system.
- Some of the responsibilities of a system administrators are enabling and disabling user accounts, preserving the home directory , setting user permissions, assigning groups to users and managing passwords.
- In a single directory, we can create 60,000 users.

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CREATE AND DELETE USER ACCOUNTS

CREATING NEW USER

- Use command $**sudo adduser username**
- Enter **password** for the new account and confirm
- Enter the details of new user and press **y** and new account is created.

DELETING USER ACCOUNT

To delete an account use the command

$ sudo userdel –r username

DISABLE AN ACCOUNT

To disable an account , remove password set on the account

$ user passwd –l username

CHANGE USER LOGINNAME and password and userid

$sudo usermod –l new_name old_name

$sudo passwd username

$sudo usermod -g new_group_id username

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ADDING USERS TO USERGROUPS

CHECK EXISTING GROUPS

$ groupmod “Press tab key twice”

HOW TO CREATE GROUP

$ sudo groupadd –g group_id group_name

TO CHECK WHETHET GROUP CREATED OR NOT

$ sudo tail /etc/group

ADD USER TO A GROUP

$ sudo usermod - a – G Groupname username

$ passwd –M user1 user2 user3 group

CHECK WHETHER USER IN GROUP

$ cat/etc/group

REMOVING USER FROM THE GROUP

$ sudo deluser username groupname

DELETE GROUP

groupdel groupname

.

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GROUP MANAGEMENT

- There are 2 categories of groups in the Linux operating system i.e. **Primary** and **Secondary** groups.
- The *Primary Group* is a group that is automatically generated while creating a user with a unique user ID simultaneously a group with ID same as the user ID is created and the user gets added to the group and becomes the first and only member of the group.
- This group is called the primary group. The *secondary group* is a group that can be created separately with the help of commands and we can then add users to it by changing the group ID of users.

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SUDO USERS

- **Add new user to the system**
  $ sudo adduser username

- **update new users password**
  $ sudo passwd username

- **Add user to the sudo group**
  $ usermod –aG sudo username

- **Test access**
  su – username

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